Determinants, Matrices and Systems: A Reference Sheet

Determinants

A determinant is a single number computed from a square array. The examples below show how it is written and evaluated. The determinant of a two by two matrix is the product of the diagonal entries minus the product of the others:

$$\det A = \begin{vmatrix} 11 & -9 \\ 1 & -9 \end{vmatrix} = -90$$

Expanding along the first row gives the determinant of a three by three matrix:

$$\det B = \begin{vmatrix} 9 & 1 & 11 \\ 2 & 13 & -7 \\ 5 & 2 & -1 \end{vmatrix} = -695$$

Scaling entries scales the terms of the determinant:

$$\begin{vmatrix} 5a & b \\ 3c & d \end{vmatrix} = 5ad - 3bc$$

The characteristic condition below is a quadratic equation in x:

$$\begin{vmatrix} x-3 & 11 \\ 1 & x+3 \end{vmatrix} = 0$$

Cramer's rule writes each unknown as a ratio of determinants:

$$x = \frac{\begin{vmatrix} 1 & -3 \\ 5 & 3 \end{vmatrix}}{\begin{vmatrix} 13 & -3 \\ -15 & 3 \end{vmatrix}}, \quad y = \frac{\begin{vmatrix} 13 & 1 \\ -15 & 5 \end{vmatrix}}{\begin{vmatrix} 13 & -3 \\ -15 & 3 \end{vmatrix}}$$

Vertical bars around an array also denote the determinant:

$$\begin{vmatrix} 11 & 7 \\ 15 & 9 \end{vmatrix} = -6$$

A row with a single non-zero entry makes the expansion short:

$$\Delta = \begin{vmatrix} 3 & -9 & 3 \\ 0 & 5 & 0 \\ 3 & 5 & 7 \end{vmatrix} = 60$$

The determinant of a triangular matrix is the product of its diagonal:

$$\det C = \begin{vmatrix} -5 & -15 & 9 \\ 0 & 5 & 9 \\ 0 & 0 & 5 \end{vmatrix} = -125$$

The determinant of a two by two matrix is the product of the diagonal entries minus the product of the others:

$$\det A = \begin{vmatrix} 7 & -1 \\ 11 & 3 \end{vmatrix} = 32$$

Expanding along the first row gives the determinant of a three by three matrix:

$$\det B = \begin{vmatrix} 2 & 1 & 7 \\ 7 & 1 & 7 \\ -3 & -7 & -9 \end{vmatrix} = -200$$

Scaling entries scales the terms of the determinant:

$$\begin{vmatrix} 9a & b \\ 3c & d \end{vmatrix} = 9ad - 3bc$$

The characteristic condition below is a quadratic equation in x:

$$\begin{vmatrix} x-7 & 9 \\ 1 & x+7 \end{vmatrix} = 0$$

Cramer's rule writes each unknown as a ratio of determinants:

$$x = \frac{\begin{vmatrix} 2 & 11 \\ 1 & -3 \end{vmatrix}}{\begin{vmatrix} 15 & 11 \\ -9 & -3 \end{vmatrix}}, \quad y = \frac{\begin{vmatrix} 15 & 2 \\ -9 & 1 \end{vmatrix}}{\begin{vmatrix} 15 & 11 \\ -9 & -3 \end{vmatrix}}$$

Matrices

Matrices record data, transformations and systems of equations. Brackets and parentheses are both common. Transposing a matrix swaps its rows and columns:

$$A = \begin{bmatrix} -7 & -7 & -4 \\ -1 & 11 & 7 \end{bmatrix}, \qquad A^T = \begin{bmatrix} -7 & -1 \\ -7 & 11 \\ -4 & 7 \end{bmatrix}$$

A matrix times a column vector gives another column vector:

$$\begin{pmatrix} 2 & -7 \\ 1 & -9 \end{pmatrix} \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$$

The inverse of a two by two matrix uses the determinant and swaps entries:

$$A^{-1} = \frac{1}{15} \begin{bmatrix} 3 & -1 \\ 9 & 2 \end{bmatrix}$$

The augmented matrix of a system keeps the right-hand side beside the coefficients:

$$\left[\begin{array}{cc|c} -5 & 0 & 1 \\ -4 & -7 & 2 \end{array}\right]$$

A three by three system gives an augmented matrix with one extra column:

$$\left[\begin{array}{ccc|c} -7 & 7 & 9 & 9 \\ 7 & 7 & 7 & 3 \\ 7 & 2 & -1 & 18 \end{array}\right]$$

Row operations turn the left block into the identity and read off the solution:

$$\left[\begin{array}{cc|c} 3 & 3 & 9 \\ 9 & 1 & 19 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right]$$

Scalar multiples of matrices are added entry by entry:

$$3\begin{bmatrix} -15 & -2 \\ 2 & 2 \end{bmatrix} + 7\begin{bmatrix} -11 & 11 \\ -5 & -4 \end{bmatrix} = \begin{bmatrix} -122 & 71 \\ -29 & -22 \end{bmatrix}$$

A vector in three dimensions is written as a column:

$$\mathbf{v} = \begin{pmatrix} 1 \\ -5 \\ 4 \end{pmatrix}$$

A scalar multiple of the identity matrix has the same number down the diagonal:

$$5I = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

Transposing a matrix swaps its rows and columns:

$$A = \begin{bmatrix} 2 & 11 & 13 \\ -7 & 9 & -15 \end{bmatrix}, \qquad A^T = \begin{bmatrix} 2 & -7 \\ 11 & 9 \\ 13 & -15 \end{bmatrix}$$

A matrix times a column vector gives another column vector:

$$\begin{pmatrix} 7 & 3 \\ -15 & 11 \end{pmatrix}\begin{pmatrix} 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 25 \\ -71 \end{pmatrix}$$

The inverse of a two by two matrix uses the determinant and swaps entries:

$$A^{-1} = \frac{1}{-59}\begin{bmatrix} 7 & -2 \\ 2 & -9 \end{bmatrix}$$

The augmented matrix of a system keeps the right-hand side beside the coefficients:

$$\left[\begin{array}{cc|c} 3 & -15 & 15 \\ 3 & -15 & 1 \end{array}\right]$$

A three by three system gives an augmented matrix with one extra column:

$$\left[\begin{array}{ccc|c} 9 & 1 & -9 & 4 \\ 15 & -5 & 9 & 7 \\ 3 & -3 & 13 & 14 \end{array}\right]$$

Row operations turn the left block into the identity and read off the solution:

$$\left[\begin{array}{cc|c} 5 & 5 & 0 \\ -1 & 1 & -4 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -2 \end{array}\right]$$

Systems of Equations

A brace on the left groups equations that must hold together. Each system below has exactly one solution. Solve the system of equations by elimination:

$$\begin{cases} -3x + 7y = 44 \\ 7x + 9y = 24 \end{cases}$$

Find every pair of numbers that satisfies the system:

$$\begin{cases} -5x + 9y = -19 \\ 11x + 5y = 17 \end{cases}$$

Use substitution to solve the following system:

$$\begin{cases} 3x + 11z = 8 \\ -5y - 9z = -24 \\ 11x - 7y - 2z = -34 \end{cases}$$

Determine the values of the unknowns in the system:

$$\begin{cases} -3x + 3y = 6 \\ 7x + 7y = -28 \end{cases}$$

Solve the system of equations by elimination:

$$\begin{cases} 7x + 11y = -29 \\ 7x - 2y = -3 \end{cases}$$

Find every pair of numbers that satisfies the system:

$$\begin{cases} -9x + 7y - 3z = 10 \\ 9x + 2y - 9z = 11 \\ 9x + 3z = -3 \end{cases}$$

Use substitution to solve the following system:

$$\begin{cases} 7x - 2y = 27 \\ 11x - 7y = 54 \end{cases}$$

Determine the values of the unknowns in the system:

$$\begin{cases} -9x + 7y = 1 \\ 11x + 5y = 53 \end{cases}$$

Solve the system of equations by elimination:

$$\begin{cases} 5x - 11y + 11z = -8 \\ 3x - 7y = -20 \\ 2x + 2z = 14 \end{cases}$$

Find every pair of numbers that satisfies the system:

$$\begin{cases} -5x + 2y = 20 \\ 9x - 9y = -63 \end{cases}$$

Worked Derivations

Several lines of an argument can share one display, aligned on the equals sign. Isolating the unknown takes three short steps:

$$\begin{aligned} 11x + 9 &= 31 \\ 11x &= 22 \\ x &= 2 \end{aligned}$$

Expanding a square by hand shows where the middle term comes from:

$$\begin{aligned} (x+3)^2 &= (x+3)(x+3) \\ &= x^2 + 3x + 3x + 9 \\ &= x^2 + 6x + 9 \end{aligned}$$

Completing the square solves the quadratic in four steps:

$$\begin{aligned} x^2 + 14x + 5 &= 0 \\ (x+7)^2 &= 44 \\ x + 7 &= \pm\sqrt{44} \\ x &= -7 \pm \sqrt{44} \end{aligned}$$

A difference of squares factors at once:

$$\begin{aligned} x^2 - 121 &= x^2 - 11^2 \\ &= (x-11)(x+11) \end{aligned}$$

The power rule is applied term by term:

$$\begin{aligned} \frac{d}{dx}\left(9x^3 + 3x\right) &= 9 \cdot 3x^2 + 3 \\ &= 27x^2 + 3 \end{aligned}$$

Adding the two equations of a system removes y:

$$\begin{aligned} 2x + y &= 4 \\ x - y &= -1 \\ 3x &= 3 \\ x &= 1 \end{aligned}$$

Matching bases turns an exponential equation into a linear one:

$$\begin{aligned} 2^{x+6} &= 512 \\ 2^{x+6} &= 2^9 \\ x + 6 &= 9 \\ x &= 3 \end{aligned}$$

A sum splits into a multiple of a known sum and a constant:

$$\begin{aligned}\sum_{k=1}^{10}(2k+3) &= 2\sum_{k=1}^{10}k + 3\cdot 10\\ &= 2\cdot 55 + 30\\ &= 140\end{aligned}$$

Isolating the unknown takes three short steps:

$$\begin{aligned}5x + 3 &= 13\\ 5x &= 10\\ x &= 2\end{aligned}$$